Chapter 6 — Heat and Thermodynamics

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Kinetic theory of gases and the gas laws

Q.1 State the basic assumptions of the kinetic theory of gases and explain the ideal gas equation.

The kinetic theory assumes that a gas consists of a large number of tiny particles in continuous random motion; the size of the particles is negligible compared with the distances between them; there are no intermolecular forces except during collisions; collisions are perfectly elastic; and the average kinetic energy of the particles is proportional to the absolute temperature. Pressure is explained as the force per unit area produced by particles colliding with the walls of the container.

The equation of state of an ideal gas is $PV = nRT$, where n is the number of moles and R is the general gas constant. It can also be written as $PV = Nk_BT$, where N is the number of molecules and $k_B = R/N_A = 1.38 \times 10^{-23}$ J K^{-1} is the Boltzmann constant. A real gas behaves most like an ideal gas at low pressure and high temperature, where the molecules are far apart and intermolecular forces are negligible.

Textbook check (p. 116): Charles' law is dated 1870, but Jacques Charles did this work in the 1780s and died in 1823. The law printed as "Joseph Lussac's law" is normally called Gay-Lussac's law.

Important Conceptual Short Questions — Topic 1

1. Why does a real gas behave like an ideal gas at low pressure and high temperature?

At low pressure the molecules are far apart, so their own volume and intermolecular forces are negligible. At high temperature the kinetic energy is large compared with attractive forces, so the gas approaches ideal behaviour.

2. How does the kinetic theory explain the pressure of a gas?

Gas particles collide with the walls and change momentum. The rate of change of momentum produces a force, and force per unit area is the pressure of the gas.

3. Why must absolute temperature be used in the gas laws?

The gas laws are proportionalities that require a scale whose zero means zero thermal motion. The kelvin scale starts at absolute zero, so ratios such as V/T and P/T are physically meaningful.

4. How do pressure, volume and temperature change in the simple gas laws?

At constant temperature, Boyle's law gives $P \propto 1/V$. At constant pressure, Charles' law gives $V \propto T$. At constant volume, Gay-Lussac's law gives $P \propto T$. In each case the third variable is kept fixed.

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Topic 2 — Internal energy, heat and work

Q.2 Explain internal energy and the first law of thermodynamics, and show that work done at constant pressure is PΔV.

Internal energy U is the sum of the kinetic and potential energies of all the molecules of a system. For an ideal gas, intermolecular potential energy is negligible, so internal energy depends only on temperature. A rise in temperature increases the average molecular kinetic energy and therefore increases internal energy.

When a gas expands at constant pressure P through a volume change ΔV, the work done by the gas is W = PΔV. This is also the area under the P–V graph. The first law of thermodynamics is Q = ΔU + W, where Q is the heat supplied to the system, ΔU is the increase in internal energy and W is the work done by the system. It is a statement of conservation of energy.

Important Conceptual Short Questions — Topic 2

1. Why does the internal energy of an ideal gas depend only on temperature?

In an ideal gas there are no intermolecular forces, so potential energy is negligible. The internal energy is therefore the total molecular kinetic energy, which is proportional to absolute temperature.

2. How is the area under a P–V graph equal to work done?

For a small expansion, work is P ΔV, which is the area of a narrow strip under the graph. Adding all strips gives the total work, so the area under the P–V curve equals the work done.

3. Why is the first law of thermodynamics a statement of conservation of energy?

It says that heat supplied to a system is used partly to increase internal energy and partly to do work. Energy is neither created nor destroyed, only transferred or converted.

4. Why do two bodies at the same temperature remain in thermal equilibrium?

Heat flows only because of a temperature difference. When the temperatures are equal there is no net heat flow between the bodies, so their thermal states do not change.

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Topic 3 — Isothermal and adiabatic processes

Q.3 Compare isothermal and adiabatic processes and apply the first law to each.

An isothermal process takes place at constant temperature. For an ideal gas, $\Delta U = 0$ because internal energy depends only on temperature. The first law then gives $Q = W$: heat supplied is completely converted into work done by the gas. The process must be slow so that heat can flow and keep the temperature constant, and PV = constant.

An adiabatic process takes place without exchange of heat, so $Q = 0$. The first law gives $\Delta U = -W$: work done by the gas comes from its internal energy, so the temperature falls during expansion and rises during compression. The process is fast or the system is insulated, and PV^{γ} = constant, where $\gamma = C_p/C_v$.

Feature	Isothermal	Adiabatic
Temperature	Constant	Changes
Heat exchange	$Q \neq 0$	$Q = 0$
Internal energy of ideal gas	$\Delta U = 0$	$\Delta U = -W$
Equation	PV = constant	PV^{γ} = constant
Speed of process	Slow	Fast or insulated

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Important Conceptual Short Questions — Topic 3

1. Why is the change in internal energy zero in an isothermal process of an ideal gas?

The internal energy of an ideal gas depends only on temperature. Since the temperature is constant in an isothermal process, the internal energy does not change.

2. Why does the temperature of a gas fall during adiabatic expansion?

No heat enters the gas, so the work done in expansion comes from internal energy. The molecular kinetic energy decreases, and therefore the temperature falls.

3. Why is an adiabatic curve steeper than an isothermal curve on a P–V graph?

In adiabatic expansion both volume and temperature change, so pressure falls more rapidly than in isothermal expansion where temperature is constant. Hence the adiabatic curve is steeper.

Topic 4 — Heat engines, second law and Carnot engine

Q.4 Explain the working of a heat engine, the second law of thermodynamics and Carnot's theorem.

A heat engine converts thermal energy into mechanical work in a cyclic process. It absorbs heat Q_1 from a hot reservoir at T_1, does work W, and rejects heat Q_2 to a cold reservoir at T_2. Its efficiency is $\eta = W/Q_1 = 1 - Q_2/Q_1$. The second law states that heat cannot be completely converted into work in a cyclic process; some heat must be rejected to a colder reservoir.

A Carnot engine is an ideal reversible engine operating between two reservoirs. Its efficiency is $\eta = 1 - T_2/T_1$, where temperatures are in kelvin. Carnot's theorem states that no engine operating between the same two temperatures can be more efficient than a Carnot engine, and all Carnot engines between the same temperatures have the same efficiency.

Textbook check (p. 122): the box credits the first practical steam engine to Braithwaite and Ericsson around 1829. Thomas Newcomen's practical steam engine dates from 1712.

Important Conceptual Short Questions — Topic 4

1. Why can no heat engine be 100% efficient?

Efficiency could reach 100% only if no heat were rejected, which would require a cold reservoir at 0 K. Such a reservoir is not available, and the second law forbids complete conversion of heat into work in a cycle.

2. How does raising the hot-reservoir temperature increase Carnot efficiency?

Carnot efficiency is $1 - T_2/T_1$. Increasing T_1 makes the ratio T_2/T_1 smaller, so a larger fraction of the absorbed heat can be converted into work.

3. Why are real engines less efficient than a Carnot engine?

Real engines have friction, turbulence and other irreversible losses. A Carnot engine is perfectly reversible, so it sets the upper limit that real engines cannot reach.

4. How does a reversible process differ from an irreversible process?

A reversible process can be retraced by an infinitesimal change in conditions, leaving no net change in the system or surroundings. An irreversible process, such as free expansion or heat flow through a finite temperature difference, cannot be exactly reversed and increases total entropy.

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Topic 5 — Refrigerator and entropy

Q.5 Explain the working of a refrigerator and the concept of entropy.

A refrigerator is a heat engine working in reverse. External work W is done on the system to extract heat Q_2 from a cold reservoir and reject heat $Q_1 = Q_2 + W$ to a hot reservoir. Its coefficient of performance is $E = Q_2/W$. It does not violate the second law because heat is moved from cold to hot only by supplying external work.

Entropy S is a state variable that measures the disorder of a system. For a reversible transfer of heat ΔQ at absolute temperature T, the entropy change is $\Delta S = \Delta Q/T$. In all natural irreversible processes the total entropy of the system plus surroundings increases. Increase in entropy means that some energy becomes unavailable for doing work.

Important Conceptual Short Questions — Topic 5

1. Why does a refrigerator not violate the second law of thermodynamics?

The second law forbids heat flowing from cold to hot without external work. A refrigerator uses a compressor to do work, so the transfer is allowed and the total entropy still increases.

2. Why does entropy increase when heat flows from a hot body to a cold body?

The cold body gains entropy Q/T_{cold} while the hot body loses Q/T_{hot}. Since T_{cold} is smaller, the gain is larger than the loss, so the net entropy change is positive.

3. How is entropy related to the availability of energy for work?

When entropy increases, energy becomes more spread out and less able to do useful work. For example, mixed hot and cold water cannot spontaneously separate again, so some energy is no longer available for work.

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